

# Phase 3 Submission Packet

## Project Title

Constraint-Aware Fitness Planner

## Team Members

Shiwei Jiang

## Selected Track

Track B: Interactive artifact

## One-Paragraph Project Summary

Constraint-Aware Fitness Planner is a local browser artifact for non-clinical diet and workout planning under real-world constraints such as symptom-trigger histories, dietary restrictions, limited equipment, schedule limits, budget limits, and adherence challenges. The system is agentic because it coordinates structured intake, ambiguity handling, safety-boundary stopping, human prioritization, reviewer-led revision, explicit packet-level handoffs, and state-preserving adaptation through structured post-acceptance execution feedback. The final artifact now delivers a clearer reviewed user plan package with a profile-fit summary, sample meal pattern, grocery focus, session blueprint, expanded weekly schedule, rationale, warnings, next-step guidance, and a downloadable plan-summary export, while keeping the coordination trace and evidence layer visible for grading.

## Repository Or Project Link

Current review package for local inspection:

- Clean local repository folder: `final_repo_ready/team-project/`

GitHub repository:

- <https://github.com/Shiwei-Jiang/multi-agents-project-constraint-aware-fitness-planner.git>

## Video Link

- [https://drive.google.com/file/d/1t9Of0rJVdyEZj3Q7Pm51GctNnXfZJSWq/view?usp=drive\\_link](https://drive.google.com/file/d/1t9Of0rJVdyEZj3Q7Pm51GctNnXfZJSWq/view?usp=drive_link)
- Local reference file: `media/demo_video_link.txt`

## Final Report

- Source: `docs/final_report.md`
- Exported PDF: `docs/final_report.pdf`
- Coverage: problem and user, architecture, implementation choices, evaluation methodology, results, failure analysis, governance, formal artifact schemas, lessons learned, and future improvements

## Architecture Diagram

- Primary Phase 3 architecture and workflow figure: docs/phase3\_architecture\_workflow.png
- Supporting/legacy architecture figures retained for continuity: docs/architecture\_diagram.svg; docs/architecture\_diagram\_compact.png
- Shows: bounded agent roles, controller-managed branches, handoff packets, stop conditions, and human-in-the-loop checkpoints

## Screenshot Index

- Screenshot index file: screenshots/screenshot\_index.md
- Coverage summary: landing screen, ambiguity checkpoint, boundary stop, human tradeoff branch, adaptation, uncertainty handling, state/history view, and evaluation/results view
- Screenshot set:
  - screenshots/01\_home.png
  - screenshots/02\_clarification.png
  - screenshots/03\_boundary\_stop.png
  - screenshots/04\_prioritization.png
  - screenshots/05\_adaptation.png
  - screenshots/06\_uncertainty.png
  - screenshots/07\_state\_and\_transitions.png
  - screenshots/08\_evaluation\_outputs.png

## Evaluation Summary

- Deterministic core scenarios: 7
- Extended adversarial, regression, and persona cases: 7
- Total evaluated cases: 14
- Full passes: 14
- Partial failures: 0
- Failures: 0
- Full-pass rate: 100%
- Repeated-run stability: 7/7

Interpretation note:

- The 14 / 14 result is intended as evidence of latest-version behavior within the project's bounded Phase 3 scope.
- It is based on internal scenario emulation, saved traces, deterministic reruns, and structured manual review rather than an external user study.
- Historical failures are still preserved in eval/failure\_log.md, so the pass table should be read together with the failure-analysis and governance sections rather than as a universal robustness claim.

Primary files:

- eval/test\_cases.csv
- eval/evaluation\_results.csv
- eval/failure\_log.md
- eval/evaluation\_scope\_note.md
- eval/version\_notes.md
- outputs/demo\_outputs/evaluation\_summary.md

- outputs/demo\_outputs/scenario\_trace\_index.md
- outputs/demo\_outputs/end\_to\_end\_trace\_baseline.md
- outputs/exported\_artifacts/automatic\_evaluation\_snapshot.json
- outputs/exported\_artifacts/stability\_check.json
- outputs/exported\_artifacts/extended\_evaluation\_snapshot.json
- docs/package\_index.md

## Submitted Files And Folders

- README.md
- AI\_USAGE.md
- platform\_notes.md
- app/
- src/
- scripts/
- docs/
- screenshots/
- eval/
- outputs/
- data/
- media/
- presentation/
- phase\_submissions/

## Notes For Final Submission Assembly

- The packet wording now avoids machine-specific filesystem paths and is ready to accept either a final GitHub URL or a compressed project-folder submission.
- The stakeholder wording in the report has been aligned with the Phase 1 feedback so the operational stakeholder is a trainer, coach, licensed dietitian, or referring provider rather than the evaluator.
- The final report now includes an explicit operational risk register and lightweight formal schemas for the main handoff artifacts and session state, which more directly addresses the Phase 2 feedback about governance structure and data clarity.
- The report PDF has been regenerated from the current Phase 3 source.
- The screenshot set has been regenerated after the latest `phase3-v1.6` artifact and evidence update.
- The evaluation exports have been regenerated from `node scripts/run_evaluation.mjs`.
- The final approved project video link is now recorded in `media/demo_video_link.txt`.